Chapter 1

Introduction to Power Electronic Systems

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Introduction

What is power electronics?

"This technology encompasses the use of electronic components, the application of circuit theory and design techniques, and the development of analytical tools toward efficient electronic conversion, control, and conditioning electric power"

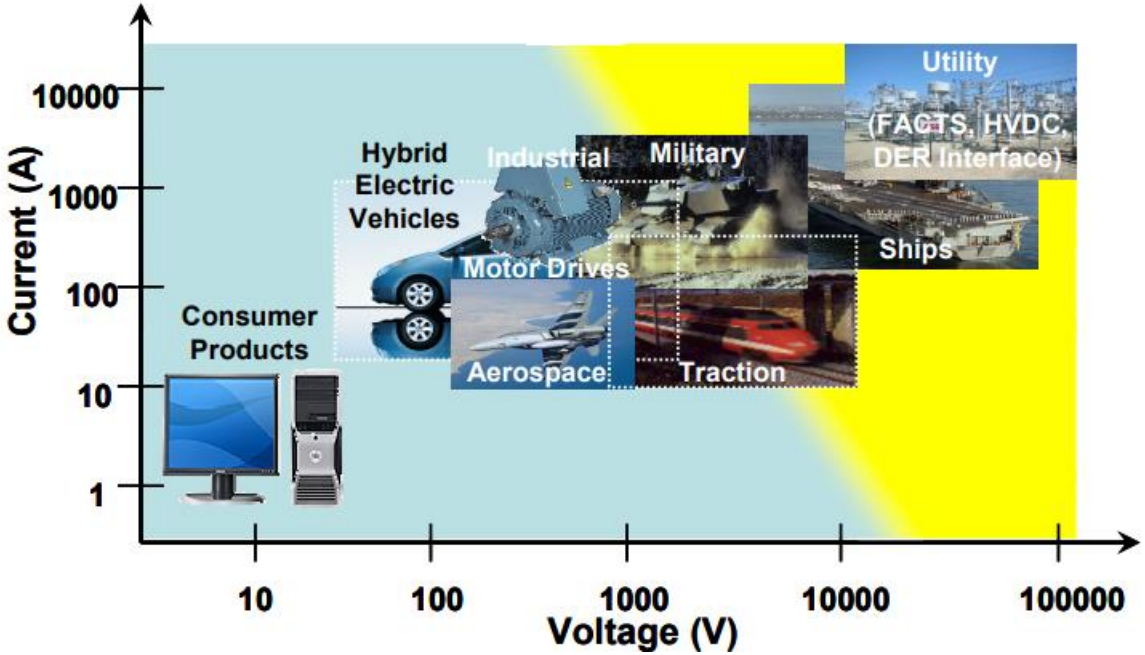
- *IEEE Power Electronics Society Definition* -

"Power electronics is the branch of electrical engineering that deals with the processing of high voltages and currents to deliver power that supports a variety of needs", "Power supply in one form is processed using power semiconductor switches and control mechanisms to another form, supplying a regulated and controlled power"

- *Synopsys website* -

Introduction

- The primary function of power electronic circuits is the processing and control of electrical energy
  - It is core technology in our electrical infrastructure
  - The majority of electrical energy used passes through one or more power electronic converters
- In many application, the power electronic circuitry is a major factor determining system size, functionality, and performance

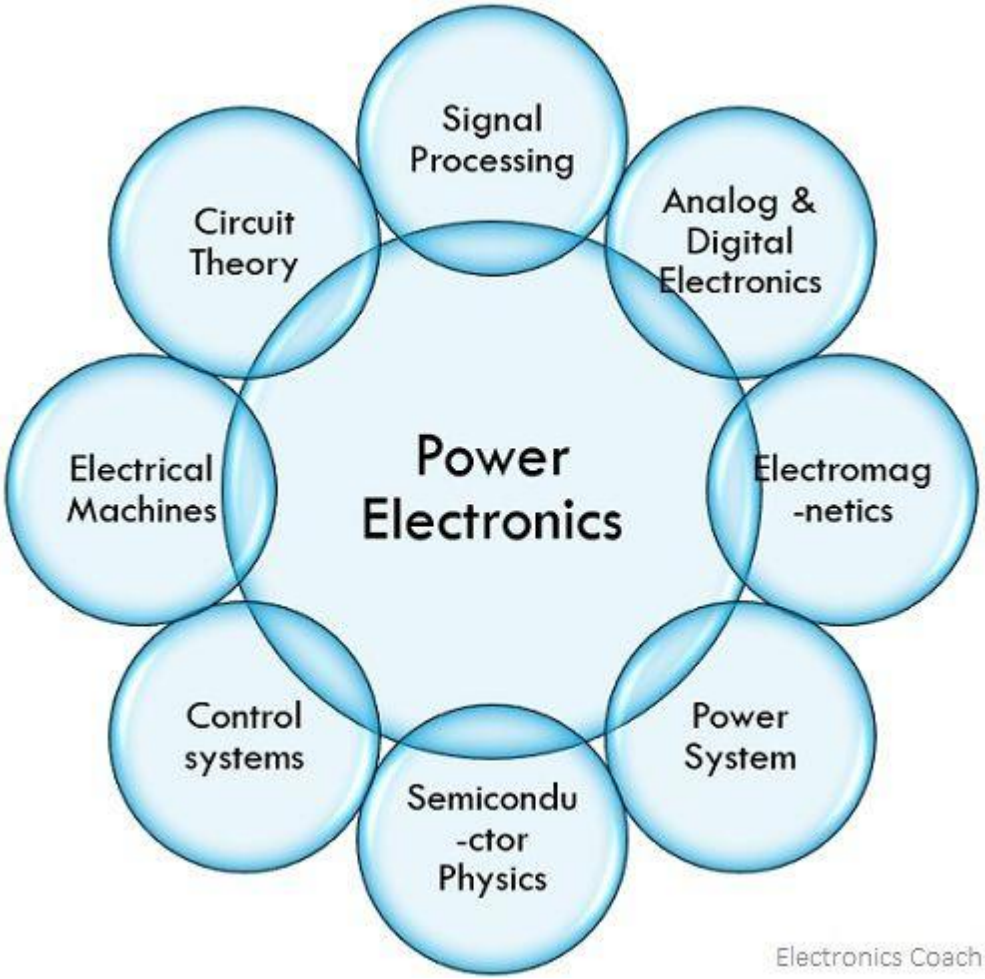


Source: Tolbert et al, "Power Electronics for Distributed Energy Systems and Transmission and Distribution Applications," ORNL 2005

Introduction

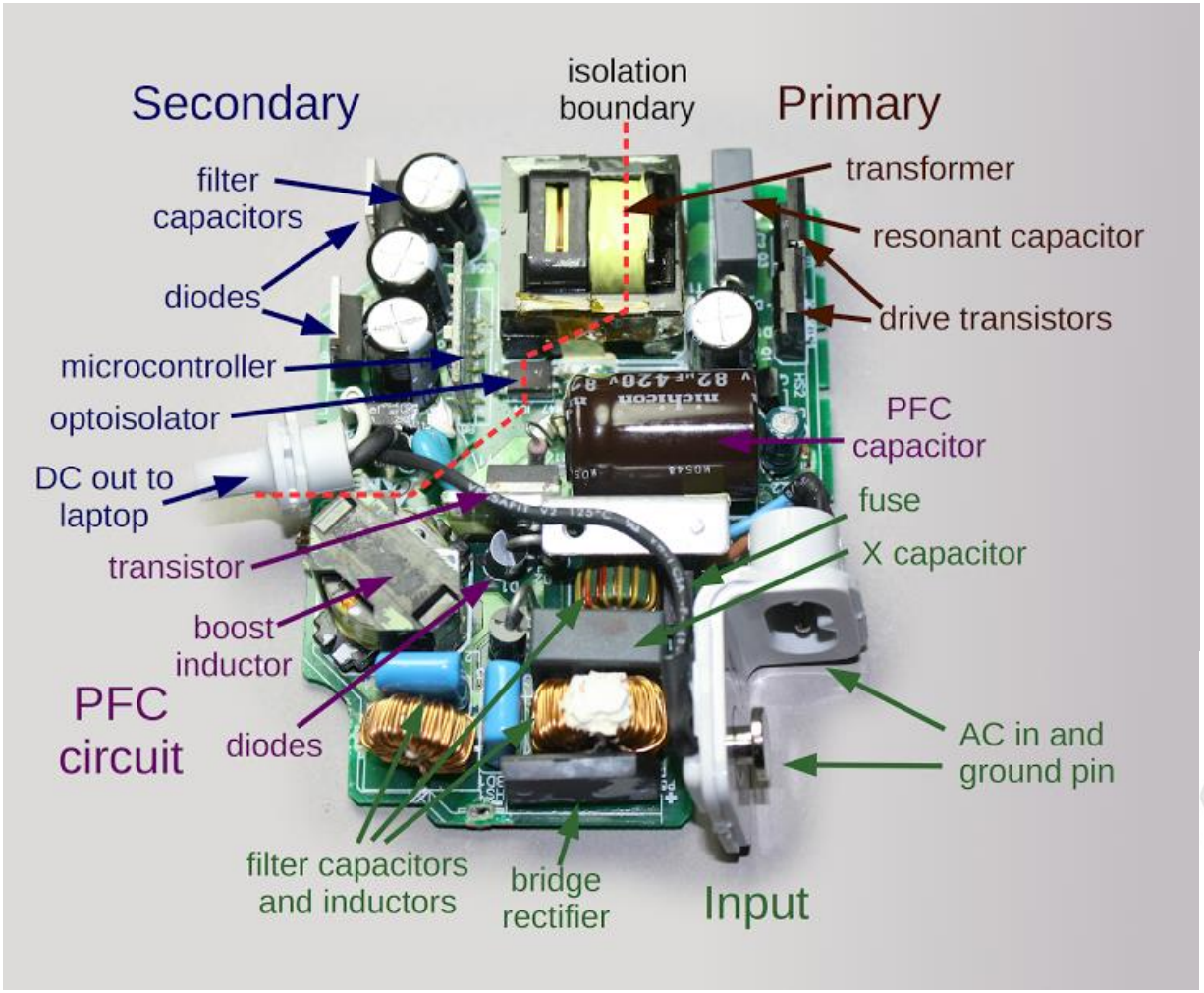
Interdisciplinary nature of power electronics

- Combining the knowledge of these diverse fields makes the study of power electronics challenging as well as interesting
- The advances in all these fields will improve the prospects for applying power electronics to new applications



Some applications of power electronic systems

Apple iphone adapter

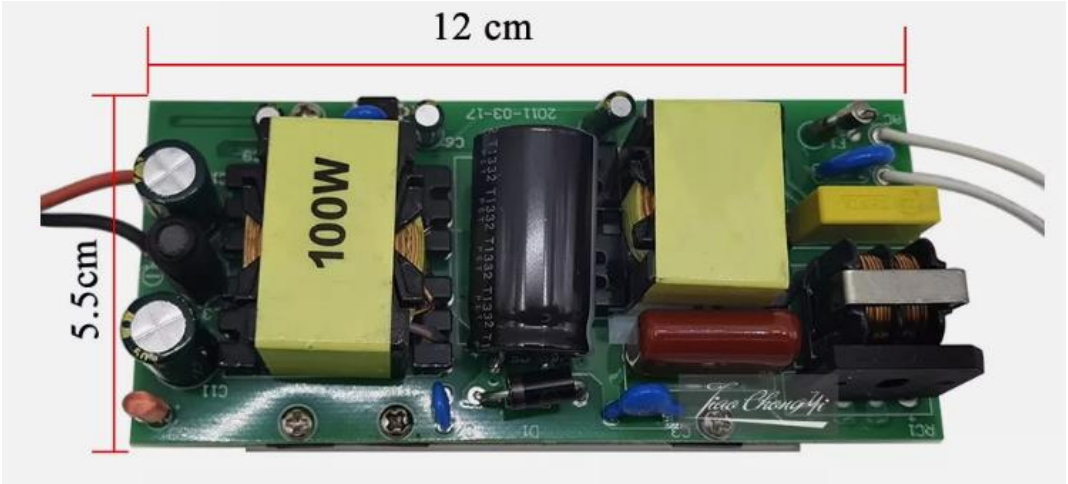


- *Frequency:* 50 to 60 Hz, single phase
- *Line Voltage:* 100 to 240 VAC
- *Output Voltage/Current:* 9 VDC/2.2 A
- *Minimum Power Output:* 20 W
- *Output Port:* USB-C



Some applications of power electronic systems

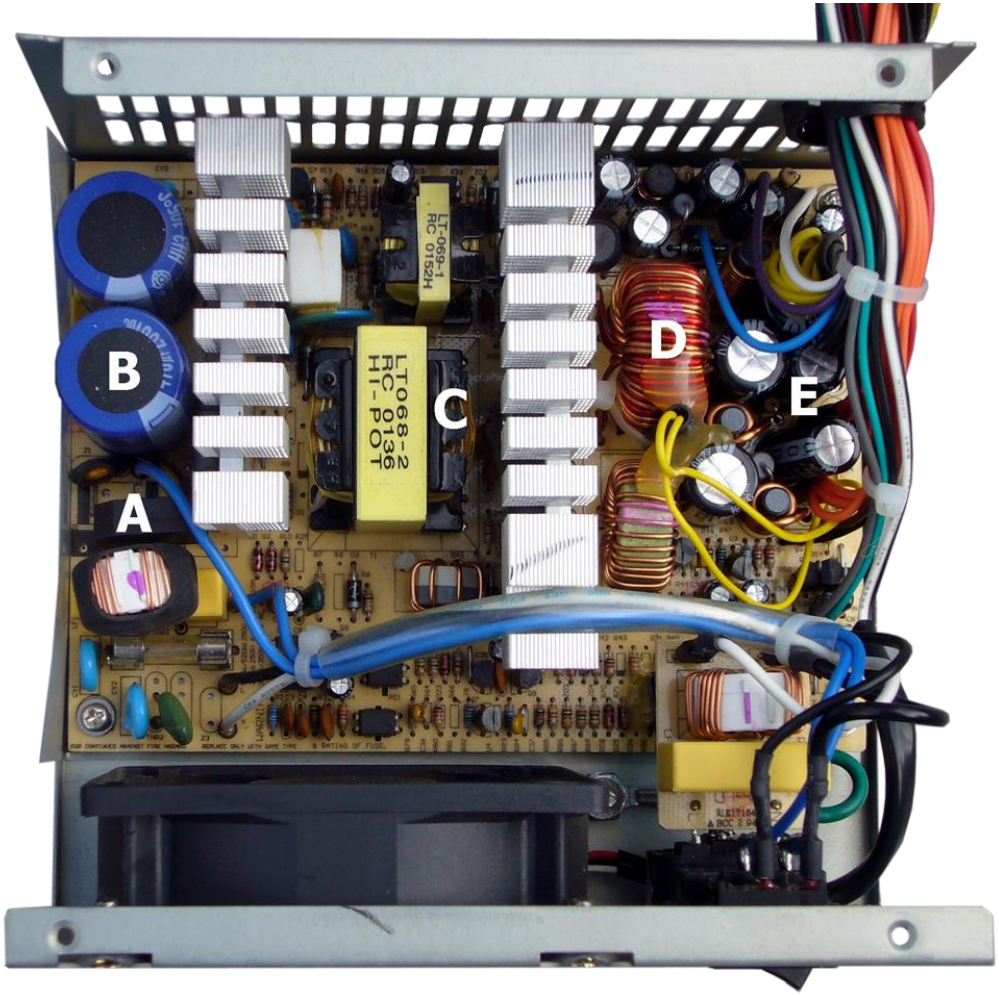
LED driver





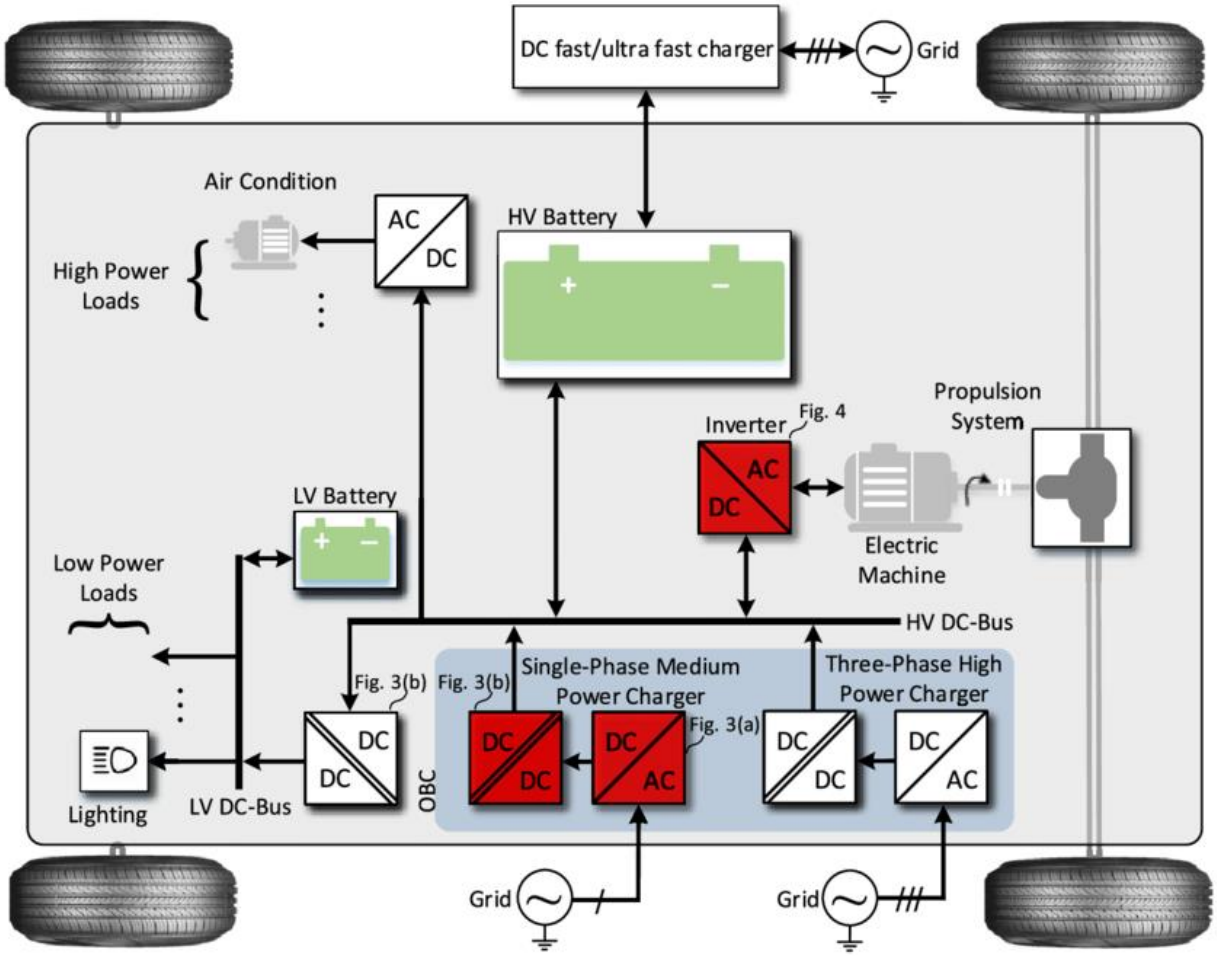
Some applications of power electronic systems

PC power supply



Some appliations of power electronic systems

Power electronics imployed in an EV

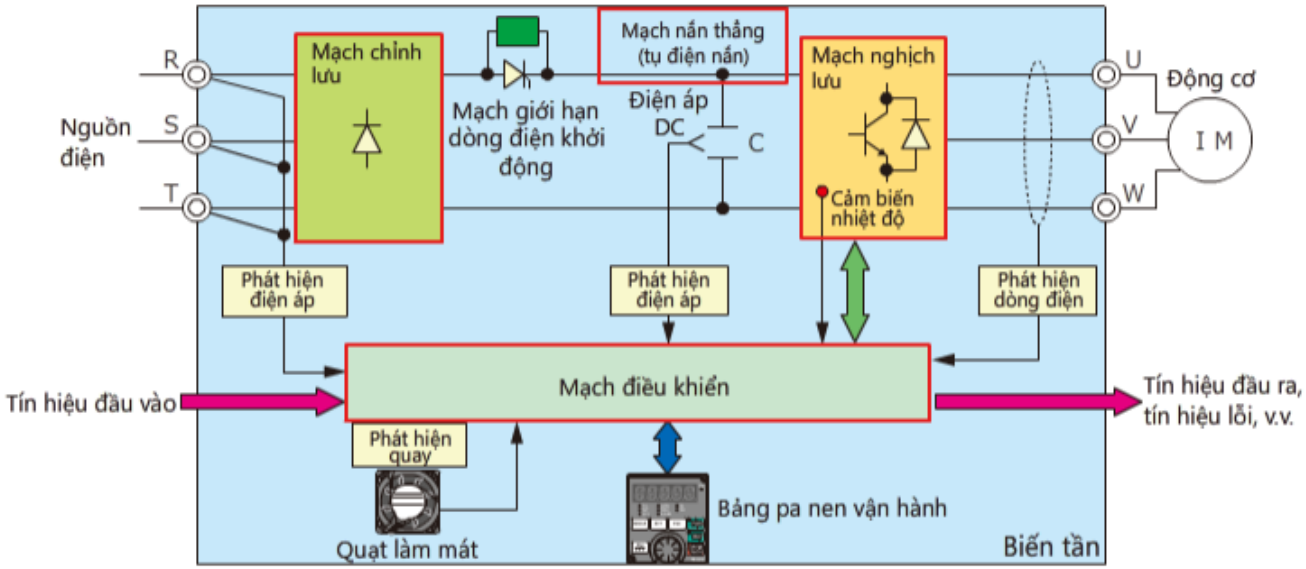


Source: Blaabjerg et al, "Reliability of power electronics systems for EV/HEV applications," Proceedings of IEEE, 2020

Some appliations of power electronic systems

Power electronics in motor drive





Tên mạch	Vai trò
Mạch chỉnh lưu	Chuyển AC sang DC.
Tụ điện nắn	Nắn điện áp DC được chuyển đổi trong Mạch chỉnh lưu
Mạch nghịch lưu	Đảo dòng DC thành AC tại tần số xác định bằng mạch điều khiển.
Mạch điều khiển	Nhận lệnh từ tín hiệu đầu vào và gửi lệnh đến Mạch nghịch lưu. Cung cấp trạng thái Mạch nghịch lưu.

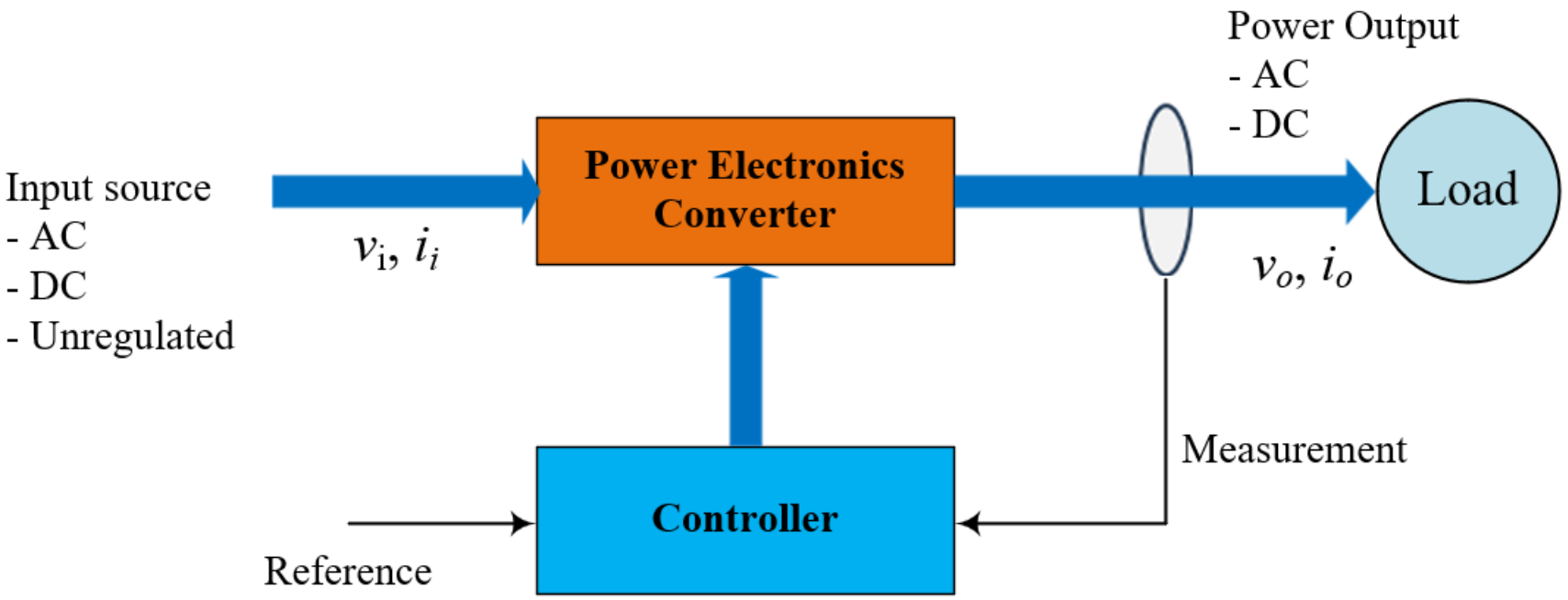
Scope of power electronic systems

- Power electronic systems can operate in the range from few watts up to GW with frequency range from some 100 Hz up to some 100 kHz, depending on the power handled

Scope of Power Electronics

Power Level (Watts)	*System*
0.1-10	• Battery-operated equipment • Flashes/strobes
10-100	• Satellite power systems • Typical offline flyback supply
100 – 1kW	• Computer power supply • Blender
1 – 10 kW	• Hot tub
10 – 100 kW	• Electric car • Eddy current braking
100 kW –1 MW	• Bus • micro-SMES
1 MW – 10 MW	• SMES
10 MW – 100 MW	• Magnetic aircraft launch • Big locomotives
100 MW – 1 GW	• Power plant
> 1 GW	• Sandy Pond substation (2.2 GW)

Structure of power electronic systems



Power electronic converter is the heart of power electronic systems

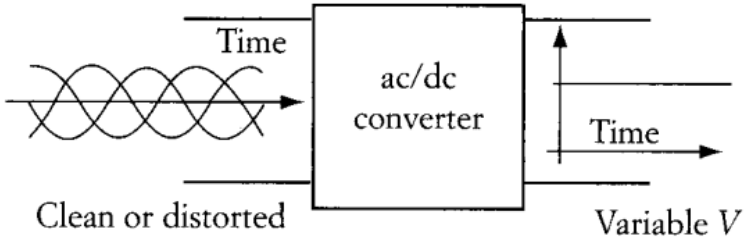
Goal of power electronic system

To convert electrical energy from one form to another, i.e., from source to load with:

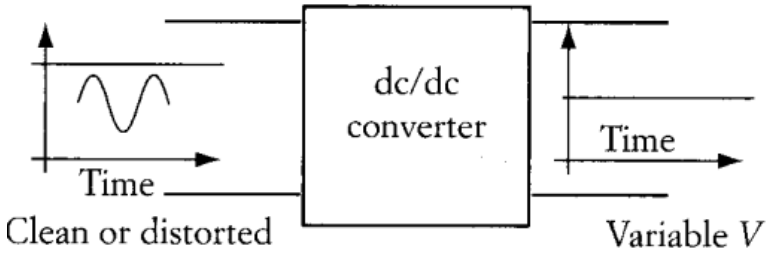
- Highest efficiency
- Highest availability
- Highest reliability
- Lowest cost
- Smallest size
- Least weight

Classification of power converter

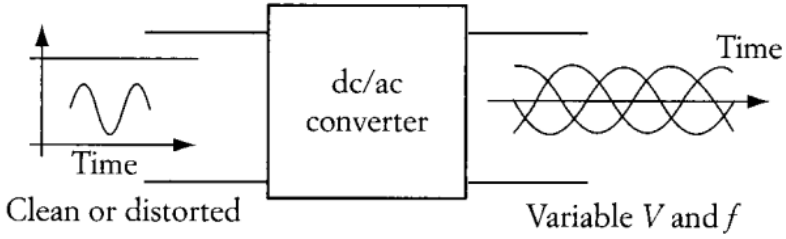
- AC to DC converter (rectifier)



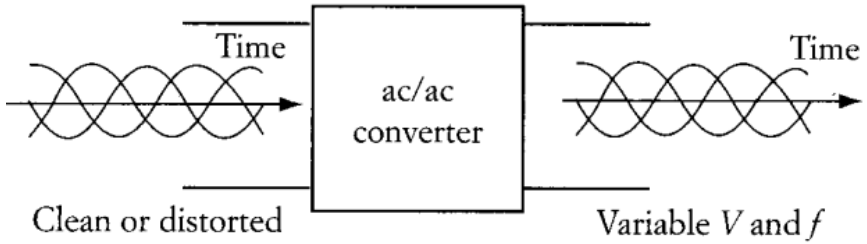
- DC to DC converter



- DC to AC converter



- AC to AC converter



Assignment

1) What is power electronics? Give at least two examples of power electronic system and explain their functions